

An Energy-Efficient and Transmission-Efficient Adaptive Routing Algorithm Using Deep Reinforcement Learning for Wireless Sensor Networks

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Abstract—How to improve energy efficiency (EE) and transmission efficiency for wireless sensor networks (WSNs) through dynamic routing is a challenge. This article introduces MADII, an adaptive routing algorithm integrating the informer model and imitation learning (IL) within a multiagent deep Q-network (DQN) framework. Our method models the routing decision in WSN as a multiagent Markov decision process (MMDP) and employs a novel sensor transmission control mechanism for information via negotiation in centralized-training, centralized-execution and centralized-distribution (STCIN3C) tailored for the centralized training and centralized execution (CTCE) paradigm. This design effectively minimizes redundant energy consumption of sensor nodes (SNs). The reward function of reinforcement learning (RL) is designed by considering transmission quantity, data-receiving success rate, and relative energy consumption. Furthermore, our approach adopts a combination of informer model and IL strategy for balancing exploration and exploitation of RL applications with large-scale action space through deep learning (DL) model and training sample, which can accelerate convergence speed, fit the dynamically changing features of WSN, and obtain better routing scheme. Extensive experiments demonstrate that MADII significantly outperforms existing methods on network lifetime and EE. Furthermore, the ablation experiment emphasizes the effectiveness of informer model and IL strategy in dealing with RL with large-scale action space.

Index Terms—Deep Q-network (DQN), large-scale action space, multiagent deep reinforcement learning (MADRL), routing, wireless sensor networks (WSNs).

I. INTRODUCTION

WIRELESS sensor network (WSN) is widely used in environmental monitoring in railroads [1], rivers [2], mines [3], and other environments that are not convenient

for humans to be involved in for a long time due to the flexibility of its autonomous networking and the low cost of deployment. Each sensor node (SN) is a source in the uplink communication, which communicates with sink node in a peer-to-peer manner, and periodically transmits the detection information to sink node [4]. Therefore, SNs in WSN are characterized by large number and fixed location in deployment, limited WSN energy, and different lifetime lengths while sink node energy is unlimited in survival, and uplink quantity is much larger than downlink data volume in communication with relatively concise routing and frequent changes. The above characteristics make WSN networking a game and overall planning process among SNs. The core issue becomes how to make adaptive routing decisions to balance energy consumption and data transmission amount, so as to improve data transmission efficiency and energy efficiency (EE), as well as prolong network lifetime.

Among the diverse solutions to the aforementioned WSN challenges, routing protocols stand out as a critical focus. Many studies analyzed and proposed different routing strategies for WSN. For instance, the energy-efficient fuzzy-based unequal clustering with a sleep scheduling (EFUCSS) protocol is proposed in [5], which applies fuzzy C-means (FCM) algorithm to form unequal clusters to balance the energy used via reducing the distance that data travels. An energy-efficient hierarchical clustering and routing technique using FCM (EHCR-FCM) for WSN is proposed in [6] to address the critical energy consumption challenge through dynamically optimizing grid/cluster formation and routing paths. A novel hybrid algorithm based on FCM and gray wolf optimizer (GWO) has been proposed [7], in which a clustering algorithm combining GWO and FCM was adopted and introduced to improve routing efficiency. Liu et al. [8] designed a cluster head (CH) election scheme GWO_WOA which combines GWO and whale optimization algorithm (WOA) to obtain energy-saving capabilities. Wilson et al. [9] developed the binarized spiking neural network optimized with the honey badger algorithm (HBA) to choose an optimal CH and achieve robust data delivery while minimizing energy consumption and extending the network lifetime. Altuwairiqi [10] proposes a novel multihop routing protocol for WSNs that enhances EE and secure quality of service (QoS) by employing an improved HBA algorithm. Ramalingam and Uthirapathy [11] proposed clustering proto-

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col based on an integrated Pelican Optimization Algorithm (POA) and sine cosine algorithm during cluster generation, which mitigate diversity challenges and ensure proper balancing. In [12], a secure and energy-aware clustering based on discrete sand cat swarm optimization (SEACDSC) by adapting sand cat swarm optimization algorithm (SCSO) was introduced to deal with the challenges of considering the security aspects of CHs when developing a CH selection strategy.

Conventional routing methodologies lack the capability to dynamically modify routing decisions in response to network performance metrics. Machine learning approaches, particularly reinforcement learning (RL) frameworks, offer adaptive solutions for evaluating path reliability and responding to fluctuating network topologies and congestion patterns. Through environmental interactions, RL mechanisms facilitate autonomous decision-making by enabling systems to optimize cumulative rewards through state–action–reward observations, ultimately identifying optimal operational strategies in specific network states [13]. As a model-free RL implementation, Q-learning operates without requiring prior environmental modeling, allowing network components to autonomously update routing strategies through continuous Q value evaluation of accessible paths [14]. The Q value metric quantitatively represents the effectiveness of routing decisions through reward-based assessments. Despite Q-learning's adaptability advantages, its computational demands escalate significantly when processing extensive state–action configurations. This becomes particularly challenging in tactical WSN where comprehensive multidimensional representations might be essential for capturing operational environment conditions. Deep RL (DRL) architectures like deep Q-learning (DQL) address these scalability constraints through neural network integration. DRL harnesses the pattern recognition capabilities of deep neural networks (DNNs) to expedite learning processes and improve decision-making efficiency in complex network environments [15]. The DQL framework implements a dual-network architecture with target network synchronization and specialized experience replay mechanisms through its deep Q-network (DQN) implementation, enabling sophisticated Q value approximations [16]. Empirical evaluations of DQN-based routing protocols demonstrate superior performance compared to conventional Q-routing systems, particularly in dynamic network environments [17]. In [18], a DQN-based energy-aware routing technique in WSN is applied for agricultural monitoring, saving residual energy to increase network lifetime. The multiagent DRL (MADRL) paradigm demonstrates particular appropriateness for mission-critical data routing applications, providing dynamic adjustment capabilities through collaborative environmental interactions and distributed experience sharing among intelligent agents [19]. The proposal in [20] implements a distributed MADRL-based routing solution for multisink tactical mobile sensor networks to overcome link layer jamming attacks.

A. Motivation

Current WSN routing and clustering research usually concentrate on addressing EE and extending lifetime of networks [21], [22]. Generally, network lifetime is closely related to

proportion of invalid SNs. Hu and Fei [23] define network lifetime in underwater sensor networks as the total routing time until the first SN dies, which normally gives a lower bound of network lifetime. Li et al. [1] define network lifetime for railway monitoring linear WSN as the number of rounds when 30% of SNs die. Zhu et al. [24] assume that the death of 85% SNs means all SNs are dead. [25] depicts network lifetime as the time duration in which the network is functional and compares network lifetime of different algorithms when all SNs die. However, these approaches often assumed single standard of network validity related to the number of active sensors.

In terms of the multiagent training framework, He et al. [26] adopt the centralized training and decentralized execution (CTDE) framework to reduce communication overhead and alleviate the curse of dimensionality. Pei et al. [27] introduce the centralized training and policy distillation for decentralized execution (CTPDE) framework based on CTDE framework to addressing these challenges of limited information access and the need for lightweight agent networks. Xu et al. [28] handle multiagent problems using CTDE framework. Most existing studies have focused on the CTDE paradigm. The main drawback is that independently trained agents treat other agents as part of the environment, thus ignoring the joint collaborative effect among multiagents. In terms of DRL training strategy, the decision-making mechanism employs an epsilon-greedy strategy [29], where SNs select optimal next-hop destinations through comparative Q value analysis, thereby maximizing long-term cumulative rewards while maintaining adaptive routing flexibility. Manfredi et al. [30] develop a DRL-based routing protocol, in which relational features are input into the DNN utilized to approximate the routing decision space. Such algorithms lack clarity in the research details of deep learning (DL) models and DRL training strategies, making it difficult to effectively respond to the dynamic changes of WSNs, especially key challenges such as slow convergence speed and the curse of dimensionality in MADRL applications with large-scale action spaces. These observations provide convincing motivation to enhance network performance by considering different network survival criteria, DL models, and DRL training strategies.

B. Contribution of This Article

Motivated by these factors, we propose multiagent deep Q-network based on informer model and imitation learning strategy (MADII) algorithm with the optimized DL model, DRL learning strategy, and transmission control mechanism under the centralized training and centralized execution (CTCE) paradigm. Our approach comprehensively considers EE and transmission efficiency. The key contributions of this article are summarized as follows.

- 1) We propose a novel MADII algorithm for routing in WSN by modeling and addressing routing decisions problem as a multiagent Markov decision process (MMDP). This algorithm achieves multiagent routing decisions in WSN through continuous optimization of long-term rewards via environmental feedback.

- 2) We design a transmission control mechanism called sensor transmission control mechanism for information via negotiation in centralized training, centralized execution, and centralized distribution (STCIN3C) based on the integration of sensor protocol for information via negotiation (SPIN) [31] and CTCE paradigm in multiagent RL (MARL). Specifically, it applies a routing centralized distribution mechanism to better achieve network load balance that further reduces energy consumption.
- 3) To address the convergence challenges in RL applications with large-scale action spaces, we implement an innovative framework integrating the informer model with a ProbSparse attention mechanism and an imitation learning (IL) strategy based on ϵ -greedy exploration. This approach dynamically adapts to WSN changes during routing decisions, reducing time complexity, memory overhead, and action space dimensionality under the CTCE paradigm. Additionally, it balances exploration–exploitation trade-offs to accelerate DRL convergence.
- 4) We formulate a new reward function for the DQN model that comprehensively considers the amount of data transmitted, data-receiving success rate, and the relative energy consumption. The design enhances both EE and transmission efficiency while extending the lifespan of WSN.

II. MODEL AND ASSUMPTIONS

A. WSN Mode

For the WSN deployed in the detection scene, this article makes the following assumptions: there are N SNs randomly distributed in a monitoring area of $M \times M$ and each of them has its own unique address that remains unchanged after being deployed [32]. These SNs, with the same initial energy E , can send their own monitoring data, their own position and remaining energy as a source node, and assume the downstream SN' data computation, data fusion, and data forwarding functions as a relay node. Sink node is located in the center of the monitoring area ($M/2, M/2$), with unlimited energy and computation capacity. All SNs can communicate directly with sink node. The main structure of WSN is shown in Fig. 1.

B. Energy Model

According to the radio energy model given in the following equation [33], the energy consumed by SNs to send l bit of data consists of two parts: the transmission loss and the power amplification loss which is exponentially related to the distance:

$$E_{TX}(l, d) = \begin{cases} lE_{elec} + l\epsilon_{fs}d^2, & d < d_0 \\ lE_{elec} + l\epsilon_{amp}d^4, & d \geq d_0. \end{cases} \quad (1)$$

As shown in the following equation, the energy consumed by SNs to receive l bit data is independent of distance:

$$E_{RX}(l) = lE_{elec} \quad (2)$$

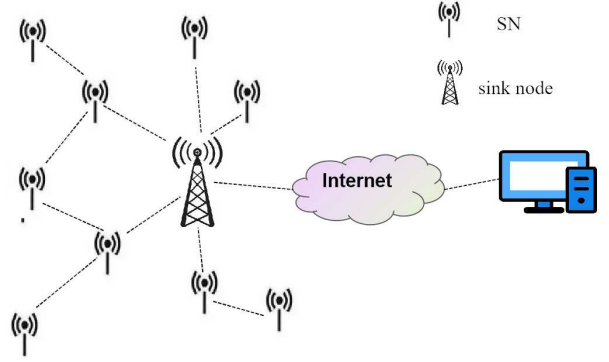


Fig. 1. WSN model.

where E_{elec} is the energy consumption for sending and receiving unit bits of data, and $d_0 = ((\epsilon_{fs}/\epsilon_{amp}))^{1/2}$ is the threshold for transmitting data between SNs. The free space channel model is applied when transmission distance is less than d_0 . When transmission distance is greater than d_0 , the multipath fading model [34] is used. ϵ_{fs} and ϵ_{amp} are the power amplification factors under the above two channel models, respectively.

III. MADII ALGORITHM

The proposed MADII algorithm shown in Fig. 2 models routing decision problem of WSN as an MMDP. MADII is an offline routing algorithm which leverages simulation data from existing routing algorithms or historical network actions paired with corresponding rewards to obtain the optimal routing selection policy. The transmission control mechanism STCIN3C is used to accomplish the routing distribution control and monitoring data transmission in WSN environment under CTCE paradigm. DQN is applied to solve optimization control problems using online measurements by interacting with environments. Informer network model can better adapt to the dynamic changes of WSN and fit Q value in corresponding to state-action pairs. Especially, combining IL strategy and informer model can reduce the action space dimension in WSN routing decisions under the CTCE paradigm through DL model and training sample, making DQN training easier to converge in large-scale action space and obtain better routing scheme.

A. DL Network Model

MADII algorithm uses informer model [35] based on ProbSparse attention mechanism to fit Q value corresponding to state-action pairs, and captures WSN features with large action space and state space through encoder-decoder structure. By calculating the Kullback-Leibler (KL) divergence between query distribution and uniform distribution, ProbSparse attention mechanism selects top- k queries to reduce the action space in routing decision process computationally, reduce the time complexity and memory cost from $O(N^2)$ to $O(N \log N)$, improve the parallel speed and efficiency in DRL training, and solve the sparse attention problem of traditional attention mechanism. This is very consistent with the characteristic that WSN only focuses on the SNs that have the greatest impact

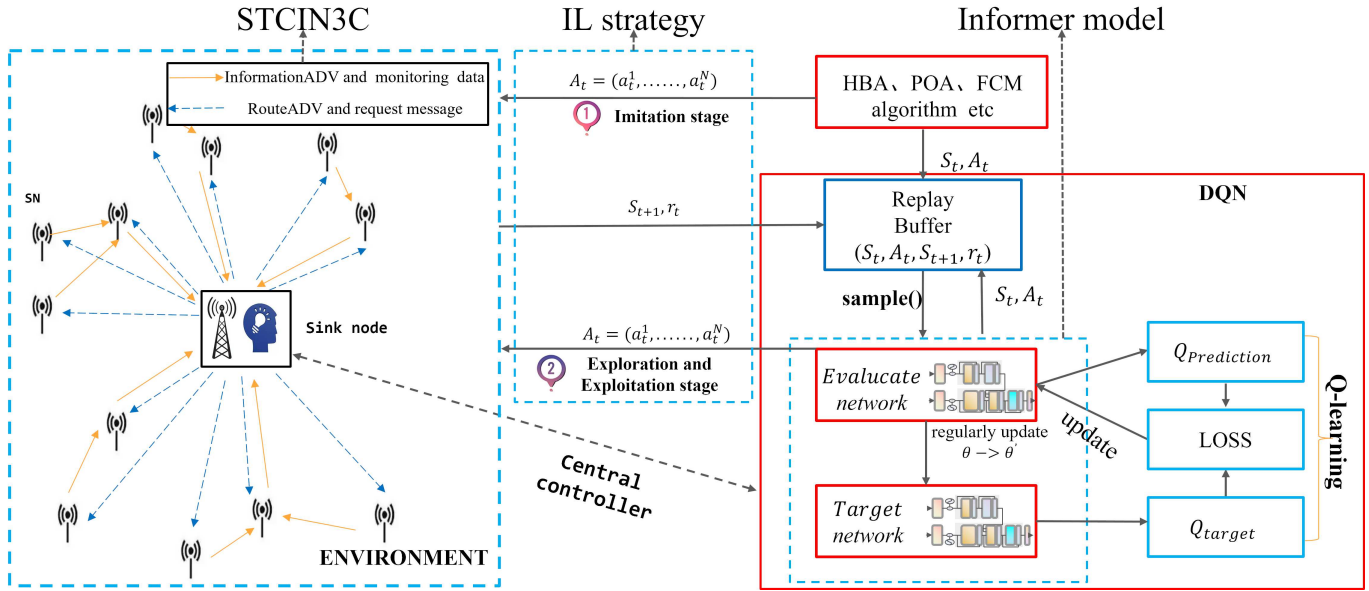


Fig. 2. MADII algorithm framework.

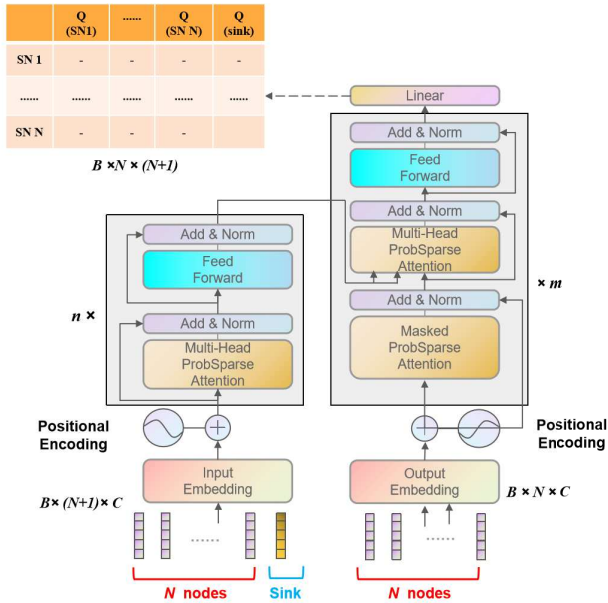


Fig. 3. Informer model.

on its routing decision. The module structure of informer is shown in Fig. 3.

1) *Preprocessing:* The low-dimensional data, including identification, position information (x, y coordinates), and residual energy, is upsampled through embedding. We use positional encoding to encode the SNs with sine-cosine position before inputting to the encoder and decoder.

2) *Encoder:* After preprocessing, the information vectors of N SNs and sink node are input into the encoder, using multihead ProbSparse attention, residual network, norm layer normalization, and feed forward network. Then the encoder learns complex features among the SNs and sink node, and

output context representations with global correlations as the input of the decoder.

3) *Decoder:* In the decode stage, the two-layer multihead ProbSparse attention uses information vectors of N SNs and the output of the encoder to predict the Q value of state-action pair. The decision ability of the model is enhanced by loss function and gradient descent optimization algorithm.

B. MARL Model

1) *STCIN3C Transmission Control Mechanism:* WSNs possess inherent communication capabilities, in which sink node has unlimited energy while SNs operate with limited energy. To enhance the overall EE and data transmission efficiency, we design a data-centric centralized transmission control mechanism named STCIN3C. This mechanism integrates the advertise-request-transfer process from SPIN, the CTCE paradigm in MARL, and the centralized routing distribution mechanism. Its primary objectives are to enable centralized route control by sink node and minimize unnecessary transmissions from SNs to the greatest extent possible.

Sink node as the central controller completes the path calculation to the target node in multiagent CTCE paradigm and establishes the routing table, where CTCE paradigm can be treated as a single-agent training model and use the shared information for training and shared information output action in the execution. Sink node will broadcast routing advertisement (actions) regularly in centralized distribution model [36]. Collecting state information is decentralized through the network itself. STCIN3C mechanism works as follows.

1) *Network Deploying:* Sink node can compute and announce the next-hop route of SNs in accordance with the planned location information and can also collect location information and compute next-hop routes in

a peer-to-peer manner. The former requires accurate deployment, and the latter will consume higher energy in the first round. This article adopts the first routing scheme after the deployment.

- 2) *Information Advertising (InformationADV) and Data Transferring*: In each subsequent round, SNs will feed-back their remaining energy, position information, and monitoring data to sink node according to the route declared by sink node in the current round.
- 3) *Routing Calculating*: Sink node calculates the next-hop route for each SN according to the collected data of all SNs by using the MADII algorithm.
- 4) *Routing Advertising (RouteADV) and Data Requesting*: The RouteADV and data request message are broadcast by sink node periodically and centralized. All SNs receive the message and update their routing tables based on the distributed routing advertisement.
- 5) *Routing Information Updating*: SNs update their routing tables only when they receive a new RouteADV.
- 6) *Routing Reissuing*: If sink node does not receive the monitoring data from a particular SN within the specified time period, it sets the next hop of that SN as sink node and broadcasts the RouteADV to the corresponding SN, and the SN in the current round sends the monitoring data directly to sink node.
- 7) *Status Confirming*: If no monitoring data is received from that SN for three consecutive times, the SN is considered dead.

2) *DQN*: DQN uses informer model to represent the state-action pair and fit Q value, introduces experience replay and target network, so as to break samples correlation, enhance the stability of the learning process, and improve sample utilization efficiency.

- 1) Based on the idea of value iteration, Q-learning [37] guides the agent to choose the best action in each state by estimating the value function Q value of each state action as shown in the following equation:

$$Q(s_t, a_t) \leftarrow (1 - \alpha) Q(s_t, a_t) + \alpha [r_{t+1} + \gamma \max_{a_{t+1}} Q_{\hat{\theta}}(s_{t+1}, a_{t+1})] \quad (3)$$

where $Q(s_t, a_t)$ is the Q value of the selected action a_t under state s_t , representing the expected cumulative rewards. r_{t+1} represents the immediate rewards obtained by selecting action a_t under state s_t . γ is the discount factor, which measures the importance of future rewards. $\max_{a_{t+1}} Q_{\hat{\theta}}(s_{t+1}, a_{t+1})$ is the Q value of selecting the optimal action under state s_{t+1} , representing the maximum expected rewards that can be obtained by selecting the optimal action from the next state. α is the learning rate, controlling the weight of new information versus old information.

- 2) DQN combines DL with Q learning to solve the problem of value function estimation in RL, the learning process of which is listed as follows.

- a) *Executing Strategy*: Select action a according to the relevant strategy. In this article, IL strategy described in Section III-E is used.

- b) *Executing Action*: Execute the action a to get the immediate reward r and a new state s' .
- c) *Storage Experience*: Put (s, a, r, s') into replay buffer D .
- d) *Extracting Sample*: Sampling (ss, aa, rr, ss') from D .
- e) *Calculation Target*: The corresponding target Q value y is calculated as follows for each extracted sample:

$$y = \begin{cases} rr, & \text{if } ss' \text{ is terminated} \\ rr + \gamma \max_{a'} Q_{\hat{\theta}}(ss', a'), & \text{otherwise.} \end{cases} \quad (4)$$

- f) *Gradient Descent*: Update network parameters using gradient descent to minimize the loss function $L(\theta)$ expressed as follows:

$$L(\theta) = \frac{1}{N} \sum_j (y_j - Q(s_j, a_j; \theta))^2 \quad (5)$$

where N is the number of samples.

- g) *Target Network Update*: The parameters $\hat{\theta}$ of the target network are updated with the parameters θ of the evaluation network after a certain number of steps.

- 3) The key elements of Markov decision process (MDP) are as follows.

- a) *Agent*: Sink node and SNs are agents. Sink node, as a central controller, has global observations (global states) to complete routing decisions for all SNs.
- b) *State Space*: The current state information SI_i of sink node and all SNs is taken as a decision factor for WSN routing, the state space S can be expressed as follows:

$$S = \{(SI_{SN1}, \dots, SI_i, \dots, SI_{SNN}, SI_{\text{sink}}) \times |SI_i = (x_i, y_i, id_i, Re_i)\} \quad (6)$$

where (x_i, y_i) is the coordinate of the node i , id_i is the identification of node i , and Re_i is the residual energy of node i .

- c) *Action Space*: Put (s, a, r, s') into replay buffer D . The action of the i th SN a_{SNi} is to select other SNs or sink node as the next hop, so the action space A can be expressed as follows:

$$A = \{(a_{SN1}, \dots, a_{SNi}, \dots, a_{SNN}) \times |a_{SNi} \in (id_{SN1}, \dots, id_{SNN}, id_{\text{sink}})\}. \quad (7)$$

- d) *Reward Function*: r_t is the reward of the agents after performing action at time t , which is covered in Section III-C.

C. Reward Function

A reward function is a formal, numerical representation of the goal of multiagent, which can be viewed as a bridge between the expected goal and the RL algorithm. Survival rate and survival duration (rounds) in WSN are important in

evaluating the routing algorithms. There would be problems if these two parameters are considered separately. Namely, the selected strategy would be that all SNs are directly linked to sink node if only the survival rate is considered. And if only the survival duration is considered, the strategy will be that the SNs closest to sink node does not take on any forwarding task and only maintains connection with sink node. Therefore, this article designs a reward function that considers the amount of data transmitted, data-receiving success rate, and the relative energy consumption. The reward function in MADII is designed as follows:

$$\begin{aligned} R_{\text{total}} &= R_{\text{valid}} * E_{\text{relCon}} * \text{Ratio}_{\text{sucFirs}} \\ &= (R_{\text{succ}} - P_{\text{fail}} - P_{\text{retry}}) * \frac{E_{\text{stand}}}{(E_{\text{start}} - E_{\text{end}})} * \\ &\quad \times \frac{(N_{\text{succ}} - N_{\text{retry}})}{N_{\text{succ}}} \end{aligned} \quad (8)$$

where $R_{\text{valid}} = R_{\text{succ}} - P_{\text{fail}} - P_{\text{retry}}$ is the final valid reward in the current cycle. R_{succ} given in the following equation represents reward for successful data transmitting from SNs to sink node, including the retransmitting part:

$$R_{\text{succ}} = \text{sum}(w_{\text{succ}} * B_{\text{succ}}) \quad (9)$$

where w_{succ} is the rewarded weight for successful data delivery and B_{succ} is given in the following equation:

$$B_{\text{succ}} = (b_{\text{succ}}^1, \dots, b_{\text{succ}}^i, \dots, b_{\text{succ}}^N) \quad (10)$$

where b_{succ}^i is the number of bits successfully sent by SN i in the current round (in kb). P_{fail} expressed in (11) represents penalty for unsuccessful data transmitting from surviving SNs to sink node. The surviving SNs may only fail to send data in the last round of survival

$$P_{\text{fail}} = \text{sum}(w_{\text{fail}} * B_{\text{fail}}) \quad (11)$$

where w_{fail} is the penalty weight for data delivering failure and B_{fail} is given in the following equation:

$$B_{\text{fail}} = (b_{\text{fail}}^1, \dots, b_{\text{fail}}^i, \dots, b_{\text{fail}}^N) \quad (12)$$

where b_{fail}^i is the number of bits unsuccessfully sent by SN i in the current round (in kb). P_{retry} expressed in the following equation represents the penalty for the surviving SNs that failed in transmitting data to sink node for the first time, then retransmitted data through the second direct connection to sink node:

$$P_{\text{retry}} = \text{sum}(w_{\text{retry}} * B_{\text{retry}}) \quad (13)$$

where $w_{\text{retry}} = 2 * w_{\text{fail}}$ and B_{retry} is given in (14). Although the retransmission strategy can guarantee the success rate of data transmission, it greatly consumes network energy and aggravates transmission delay. The penalty weight of data retransmission is twice the penalty weight of data failure

$$B_{\text{retry}} = (b_{\text{retry}}^1, \dots, b_{\text{retry}}^i, \dots, b_{\text{retry}}^N). \quad (14)$$

b_{retry}^i in (14) is the number of bits that SN i retransmits during data transmission in the current round (in kb).

R_{succ} , P_{fail} , and P_{retry} are balanced by two weights w_{fail} and w_{succ} . When $w_{\text{fail}} < w_{\text{succ}}$, the training process tends to try

different actions in order to achieve higher data transmission success rate. Otherwise, the training process tends to take conservative and reliable actions to avoid data transmission failure. In this article, we set $w_{\text{succ}} = w_{\text{fail}}$.

E_{relCon} given in the following equation indicates relative energy consumption:

$$E_{\text{relCon}} = \frac{E_{\text{stand}}}{(E_{\text{start}} - E_{\text{end}})} \quad (15)$$

where E_{start} is the total energy of all surviving SNs in the entire WSN at the beginning of the round, E_{end} is the total energy of all surviving SNs in the entire WSN at the end of the round, and E_{stand} is the total energy consumption in the current round if all surviving SNs in the entire WSN directly connect to sink node for data transmission. In WSN, SNs that are farther away from sink node consume much more energy to send an equal amount of data compared to closer SNs. And if energy consumption is used for efficiency assessment, SNs that are farther away from sink node are always more inefficient. Therefore, it is necessary to standardize the energy consumption of all the SNs to effectively assess SNs in different locations.

$\text{Ratio}_{\text{sucFirs}}$ expressed in the following equation indicates one-time transmission success rate:

$$\text{Ratio}_{\text{sucFirs}} = \frac{(N_{\text{succ}} - N_{\text{retry}})}{N_{\text{succ}}} \quad (16)$$

where N_{succ} given in (17) represents the amount of data sent successfully by all SNs of the entire WSN in the current round. N_{retry} given in (18) represents the amount of successfully retransmitted data in the entire WSN in the current round

$$N_{\text{succ}} = \text{sum}(B_{\text{succ}}) \quad (17)$$

$$N_{\text{retry}} = \text{sum}(B_{\text{retry}}). \quad (18)$$

In terms of the combination of result-oriented and process-oriented aspects, the scheme guided by reward function could transmit more data with minimal energy consumption in the current round, so that it conserves more residual energy for a longer network lifetime.

D. Performance Evaluation Metrics

1) *Survival Rounds*: Network lifetime is evaluated by survival rounds (SR) $\text{SR}(\beta)$ given in the following equation, which is defined as the number of rounds when the proportion of dead SNs reaches the expected level:

$$\begin{aligned} \text{SR}(\beta) &= \arg \max_r \left(\left(\frac{\left(\sum_{i=0}^N 1_{\{E_i(r) \leq 0\}} \right)}{N} \right) < \beta \right) \\ 0 &< \beta \leq 1 \end{aligned} \quad (19)$$

where $E_i(r)$ represents the energy of SN i in round r , N is the total number of SNs, and network survival criterion β is the proportion of dead SNs required in advance according to different scenarios.

2) *Reward Score*: Reward score (RS) is defined as follows:

$$RS = \sum_{r \in SR(\beta)} R_{total}^r \quad (20)$$

where R_{total}^r is the reward for round r . It reflects the comprehensive impact of routing schemes on EE and transmission efficiency.

3) *Data Delivery Volume (kb)*: Data delivery volume (DDV) is defined as follows:

$$DDV = \sum_{r \in SR(\beta)} \sum_{i \in N} D_i^r \quad (21)$$

where D_i^r indicates data volume sent successfully by SN i in round r .

4) *Energy Efficiency (kb/J)*: EE is defined as follows:

$$EE = \frac{DDV}{E_0 - E_{SR(\beta)}} \quad (22)$$

where E_0 represents the total energy of all SNs in the initial state of the WSN, $E_{SR(\beta)} = \sum_{i \in N} e_{SR(\beta)}^i$ represents the total energy of all SNs in the WSN after round $SR(\beta)$, $e_{SR(\beta)}^i$ is the energy of SN i after round $SR(\beta)$.

E. DRL Training Strategy

1) *Epsilon-Greedy Strategy*: Epsilon-greedy strategy, a strategy to balance the relationship between exploration and exploitation, has been widely used in the field of RL. Action a is determined by (23). The strategy is inefficient and difficult to converge in RL of large-scale action space

$$a = \begin{cases} \operatorname{argmax}_{a'} Q_{\theta}(s, a'), & w.p.(1 - \epsilon) \\ (\text{uniformly}) \text{ random action,} & w.p.\epsilon. \end{cases} \quad (23)$$

2) *IL Strategy*: In WSN, as the number of SNs and the initial energy increases, the state space—especially the action space of Q-learning—grows exponentially, and the probability of useful training samples being randomized decreases dramatically. As a result, effectively balancing the Exploration and Exploitation becomes more difficult, thus exacerbating the difficulty of training CTCE paradigm.

Therefore, based on the existing epsilon-greedy strategy, this article applied IL strategy shown in Algorithm 1 by exploiting the advantage of DQN experience replay. This learning method stores the transition (s, a, r, s') of the existing routing algorithms in the replay buffer D , then provides it to the central controller for IL, reducing the action space in routing decision process for training samples, accelerating the training speed in CTCE paradigm and further balancing exploration of the unknown strategy and exploitation of the mode already been learned. During the learning process, IL strategy could prevent inefficient exploration and speed up the training, while epsilon-greedy strategy is used in the last learning stage to avoid getting stuck in a local optimum.

As we can see from Fig. 2, the entire training model of IL strategy consists of two stages. The first one belongs to imitation stage. Algorithms such as FCM, HBA, and POA are executed to select the CHs and each SN selects the corresponding action to get the reward and puts (s, a, r, s')

Algorithm 1 DQN Based on IL Strategy

Input: number of SNs N , network survival criterion β , state space S , action space A , discount rate γ , learning rate α , Exploration probability ϵ , number of times for exploration and exploitation C_e , number of times for imitation learning C_i , reward weighting of data sending success w_{succ} , penalty weighting of data sending failure w_{fail} , penalty weighting of data sending retry success w_{retry} .

Output: Q Network $Q_{\theta}(s, a)$

Step 1 Imitation stage

Initialize the Replay Buffer D with capacity N , the start state $s \leftarrow S_{init}$, ϵ , α , m .

Load the Evaluate network and Target network with the parameter θ_{load} .

Empty the initialization Replay Buffer D .

while $m < \beta$ **do**

Execute FCM/HBA/POA etc. algorithms in state s to pick cluster heads and select actions a .

Execute the action a to get the immediate reward r and a new state s' . Storage experience.

$s \leftarrow s', m \leftarrow \frac{\sum_{i=0}^N 1_{\{E_i(r) \leq 0\}}}{N}$.

end while

for $c = 0$ C_i **do**

Extract sample and calculate target according to (4).

Perform gradient descent according to (5).

Update Target network $\hat{\theta} \leftarrow \theta$.

end for

Step 2 Exploration and Exploitation stage

for $c = 0$ C_e **do**

Initialization start state $s \leftarrow S_{init}$, ϵ , α , m , clear initialization Replay Buffer D .

while $m < \beta$ **do**

select action a according to (23).

Execute action a Storage Experience. $s \leftarrow s', m \leftarrow \frac{\sum_{i=0}^N 1_{\{E_i(r) \leq 0\}}}{N}$.

Extract sample and calculate target according to (4).

Perform gradient descent according to (5).

Update Target network $\hat{\theta} \leftarrow \theta$.

end while

end for

into replay buffer D to provide to the central controller for IL. The other is exploration and exploitation stage. After gaining a strategy with better performance in the first stage, exploration and exploitation are carried out again according to epsilon-greedy strategy to validate the learning results of it.

IV. EXPERIMENTAL SETUP AND PERFORMANCE ANALYSIS

A. Experimental Details

1) *Baselines*: This article has selected seven WSN routing methods as comparison, including FCM [5], GWO_WOA [8], HBA [9], POA [11], SCSO [12], multiagent deep Q-network base on Transformer model [38] and IL strategy (MADTI), and multiagent deep Q-network base on informer model [35] and epsilon-greedy strategy (MADIE). FCM is a fuzzy clustering algorithm. SCSO, HBA, POA, and GWO_WOA are

TABLE I
WSN AND ENERGY MODEL PARAMETERS

Parameters	Value
Length of monitoring area L/m	500
Width of monitoring area W/m	500
sink node position/ (m, m)	(250, 250)
Number of SNs	100
Packet size/ $kbit$	4
Control packet size/ $kbit$	0.1
Initial energy/ J	0.5
$E_{elec}/(nJ * bit^{(-1)})$	50
$\epsilon_{fs}/(pJ * (bit * m^2)^{-1})$	10
$\epsilon_{amp}/(pJ * (bit * m^4)^{-1})$	0.0013
Clustering number	10

population-based algorithms and solve complex optimization problems according to the fitness function. Equation (24) is the fitness function based on energy factor and position equilibrium factor [39], which is selected in experiments of this section. By comparing the performance of MADTI and MADIE belonging to MADRL algorithm, ablation experiments are conducted to study the impact of informer model and IL strategy on the algorithm performance

$$F_1 = a \frac{\frac{1}{K} \sum_{i=1}^K E_{CH}^r(i)}{\frac{1}{N-K} \sum_{j=1}^{N-K} E_{NCH}^r(j)} + (1-a) \frac{\sum_{i=1}^{N-K} d(NCH_i, \text{sink})}{\sum_{j=1}^K d(CH_j, \text{sink}) + \sum_{i=1}^{N-K} d(NCH_i, CH_j)}. \quad (24)$$

In (24), $E_{CH}^r(i)$ represents the residual energy of candidate CH SN CH_i in the r th round, $E_{NCH}^r(j)$ represents the residual energy of non-CH SN NCH_j in the r th round, $d(NCH_i, \text{sink})$ represents the distance between non-CH SN NCH_i and sink node, $d(CH_j, \text{sink})$ represents the distance between CH SN CH_j and sink node, and $d(NCH_i, CH_j)$ represents the distance from non-CH SN NCH_i to its corresponding candidate CH SN CH_j .

2) *Setup*: The basic experimental setup and platform are as follows:

- 1) When half of the SNs in the WSN die, it will lead to incomplete data collection, network communication interruption, and sharp decline in network performance. This article focuses on the routing schemes with network survival criteria of $\beta \leq 50\%$, including first node died (FND) and half node died (HND).
- 2) 100 SNs are randomly deployed in a 500×500 m area. This article uses SR, RS, EE, and DDV to evaluate the routing algorithm.
- 3) The number of training times in epsilon-greedy strategy is equal to the total number of training times in the two stages of IL strategy.
- 4) Data compression is not considered when monitoring data is forwarded.
- 5) *Platform*: All the models were trained/tested on a single NVIDIA RTX 4070 Ti 12 GB GPU.

TABLE II
INFORMER MODEL PARAMETER

Parameters	Value
e_layers	2
d_layers	1
d_ff	4096
factor	5
Dropout	0.1
Activation	gelu
batch_size	64
learning_rate	0.0001
d_model	1024

TABLE III
DQN PARAMETERS

Parameters	Value
$max_epsilon$	1
$min_epsilon$	0.1
γ (discount factor)	0.99
w_{succ}	10
w_{fail}	10
update intervals C_1 for network parameter θ	10

TABLE IV
DRL IL STRATEGY PARAMETERS

Training strategy	Training stage	Parameters	Value
IL strategy	Imitation stage	Clustering number of FCM	3-18
	Exploration and Exploitation stage	Times of Exploring and Utilizing C_e	1000

TABLE V
OPTIMIZATION ALGORITHMS PARAMETERS

Parameters	Value
Maximum number of iterations	800
Parameters a of the fitness function	0.54

3) *Main Parameters*: The parameters in the experiments are listed in the following tables. Table I presents parameters in WSN and energy model. Table II presents informer model parameter. Table III presents DQN parameters. Table IV presents parameters for RL training strategy. Table V presents parameters of optimization algorithms used for comparison.

B. Results and Analysis

1) *Comparison of Different Algorithms*: Table VI summarizes the evaluation results of all the methods for WSN

TABLE VI
EVALUATION RESULTS OF DIFFERENT ALGORITHMS

NSC β	Method	FCM	SCSO	HBA	POA	GWO_WOA	MADTI	MADIE	MADII(our method)	
	Metric	Value								Improvement over Best Baseline
1% (FND)	<i>RS</i>	32340	3457	15747	14067	20989	11853	10994	44788	38.49%
	<i>EE(kbit/J)</i>	555	289	391	379	420	352	356	817	47.21%
	<i>DDV(kbit)</i>	7596	1596	5596	5188	6792	4792	4396	7596	0.00%
	<i>SR</i>	19	4	14	13	17	12	11	19	0.00%
10%	<i>RS</i>	42754	19809	29082	27468	34013	16731	16724	78269	83.07%
	<i>EE(kbit/J)</i>	533	361	438	426	456	372	371	833	56.29%
	<i>DDV(kbit)</i>	10636	7748	9728	9664	10208	6664	6664	14216	33.66%
	<i>SR</i>	27	20	25	25	26	17	17	37	37.04%
25%	<i>RS</i>	47494	33227	44937	39692	52637	25894	25883	87928	67.05%
	<i>EE(kbit/J)</i>	531	427	498	473	517	418	418	723	36.16%
	<i>DDV(kbit)</i>	12228	12008	15208	13620	15784	10308	10308	19244	21.92%
	<i>SR</i>	32	33	42	37	43	28	28	52	20.93%
50% (HND)	<i>RS</i>	68413	63134	77117	77610	84531	49645	49453	92653	9.61%
	<i>EE(kbit/J)</i>	597	580	652	653	659	580	580	742	12.59%
	<i>DDV(kbit)</i>	18704	22480	25932	25916	26256	19692	19692	25708	-2.09%
	<i>SR</i>	57	75	87	87	86	67	67	89	2.30%
Count		2	0	0	0	1	0	0	15	

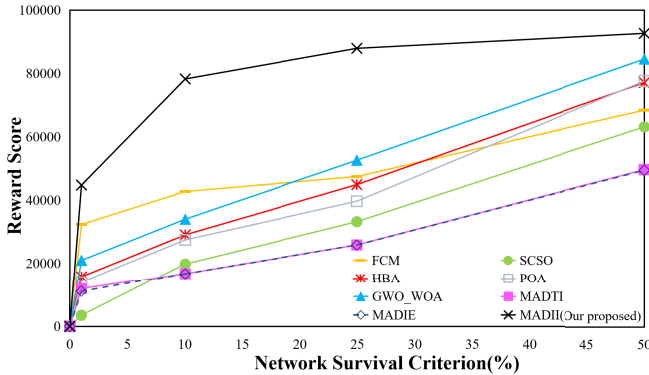


Fig. 4. Comparisons in terms of RS.

routing with specific network survival criterion. The best results are highlighted in boldface. The proposed algorithm MADII has better networking performance than the other seven algorithms. For all network survival criteria, RS, EE, and SR are better than the other seven algorithms, reflecting the overall EE and transmission efficiency of the algorithm. The percentage improvement of MADII over the best-performing algorithm among others is shown in the last column.

- 1) *RS*: As it is shown in Fig. 4, the proposed MADII algorithm gains the highest RS in groups with different network survival criteria. Compared to MADTI and MADIE as ablation study, the average scores of MADII increased by 191% and 242% in four network survival criteria. This indicates that IL strategy can significantly accelerate the convergence speed and performance of

DQN algorithms by drawing on useful experiences from various clustering algorithms. Theoretically, learning more clustering algorithms can increase the diversity of training samples. Informer model retains only the top- k important attention weights when calculating ProbSparse self-attention. When making decisions for the next hop of SNs, only the ones within a certain range need to be considered. Combining informer model and IL strategy can obviously reduce the action space in WSN routing decisions when compared with using alone. This enhances exploration efficiency so that it can achieve faster RL convergence in large-scale action space. The score gap between MADII and GWO_WOA becomes smaller as β grows, showing that the training difficulty of better routing scheme increases as the network survival criterion improves.

- 2) *EE*: As it is shown in Fig. 5, MADII outperforms the other seven algorithms for all network survival criteria with a decreasing trend. When $\beta = 1\%$ (FND), the curve is the steepest and the routing performance is the highest. As the network survival criterion gradually improves, the steepness of the curve changes slows down. The steeper EE indicates that the learned routing scheme is efficient. The trend of curve reflects that the network is more stable in the beginning of WSN, meaning all SNs have enough energy and there will not be any sudden death of SNs. As the number of dead SNs increases, SNs with moderate location from sink node are more likely to die and energy holes will appear, which has a great impact on the whole WSN and makes it more difficult to train.

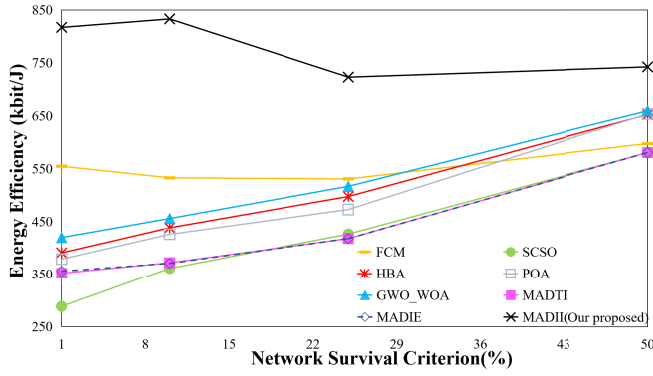


Fig. 5. Comparisons in terms of EE.

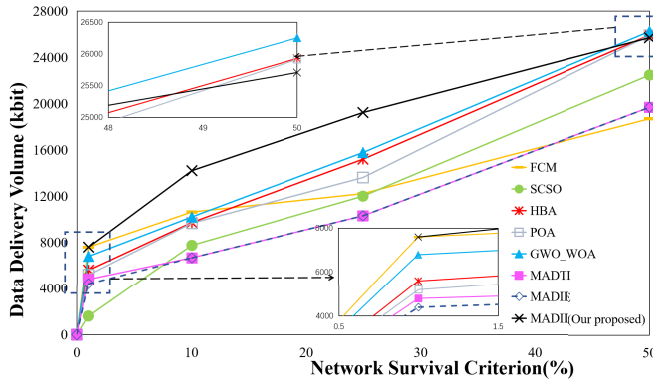


Fig. 6. Comparisons in terms of DDV.

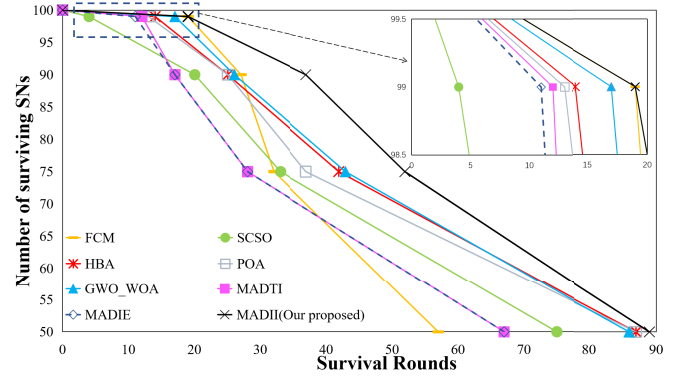


Fig. 7. Comparisons in terms of SR.

3) *DDV*: As it is shown in Fig. 6, when $\beta = 1\%$ (FND), the MADII and FCM algorithms achieve the highest successfully transmitted data volume among all comparative methods. At $\beta = 50\%$ (HND), MADII's successfully transmitted data volume is slightly lower than that of GWO_WOA, HBA, and POA but higher than that of the other comparative algorithms. RS and EE of MADII are higher. This indicates that MADII pays more attention to EE, which will in turn increase the amount of data transmitted per unit of relative energy consumption. Meanwhile, it suggests that the current routing scheme trained with $\beta = 50\%$ (HND) requires further optimization.

4) *SR*: As it is shown in Fig. 7, the proposed MADII algorithm performs better in SR than all the other methods. It indicates that the proposed reward function can not only promote the learning routing scheme with high EE but also improve the survivability of WSN.

2) *MADII Routing Scheme Analysis*: To illustrate the characteristics of the routing schemes generated by the proposed MADII algorithm under different network survival criteria, we conducted the following analysis:

1) *Routing Process*: Considering that the death of the first SN has the greatest impact on the performance and lifetime of the WSN, we can see the difference between the MADII routing scheme and traditional ones by analyzing the routing scheme when $\beta = 1\%$ (FND). By analyzing the procedure of routing schemes changing shown in Fig. 8, we can observe that the following holds.

a) The MADII routing scheme is a combination of different lifetime policies with different next-hop policies. As it is shown in Fig. 8, the routing scheme adopts the subcluster combining direct connection policy and could be roughly divided into three phases: efficient networking phase during rounds 1–4 [shown in Fig. 8(a)–(c)], energy balancing phase during rounds 5–13 [shown in Fig. 8(d)–(f)], and the sprinting phase during rounds 14–19 [shown in Fig. 8(g)–(i)].

b) In the first phase that all SNs have enough energy, SNs far away from sink node will pick an SN in moderate distance as CH or the next hop, while SNs near sink node tend to connect with it directly without relaying through the next hop. This can not only reduce the pressure of the CH or the next hop but also save the overall energy consumption of the WSN. In the second phase, as energy decline of the relay SNs becomes larger, some SNs will replace the next hop to delay the emergence of the energy black hole (in this article, you can see from the 7th to 13th round of graphs that the training is not sufficient, and most of SNs are directly connected to sink node). In the third phase, as the energy of SNs at a greater distance from sink node decreases, they will replace the next hop to the current suitable transit SNs to survive longer, which is similar to the first phase of the network. As shown in Fig. 8(i), the circled SNs did not replace the next hop in the second phase, so there would be data retransmission when an energy black hole was about to emerge. It can also be concluded from the graph that the learned routing scheme would pick a node closer as the next hop if sink node is more distant, which is reasonable from the point of view of energy saving in a single round.

2) Routing scheme comparison of different network survival criteria β . Fig. 9 shows that when $\beta = 1\%$ (FND), the curve is the steepest and routing performance is the highest. As the network survival criterion gradually improves, the steepness of the curve is slowing down, indicating that the network survival criteria focusing on

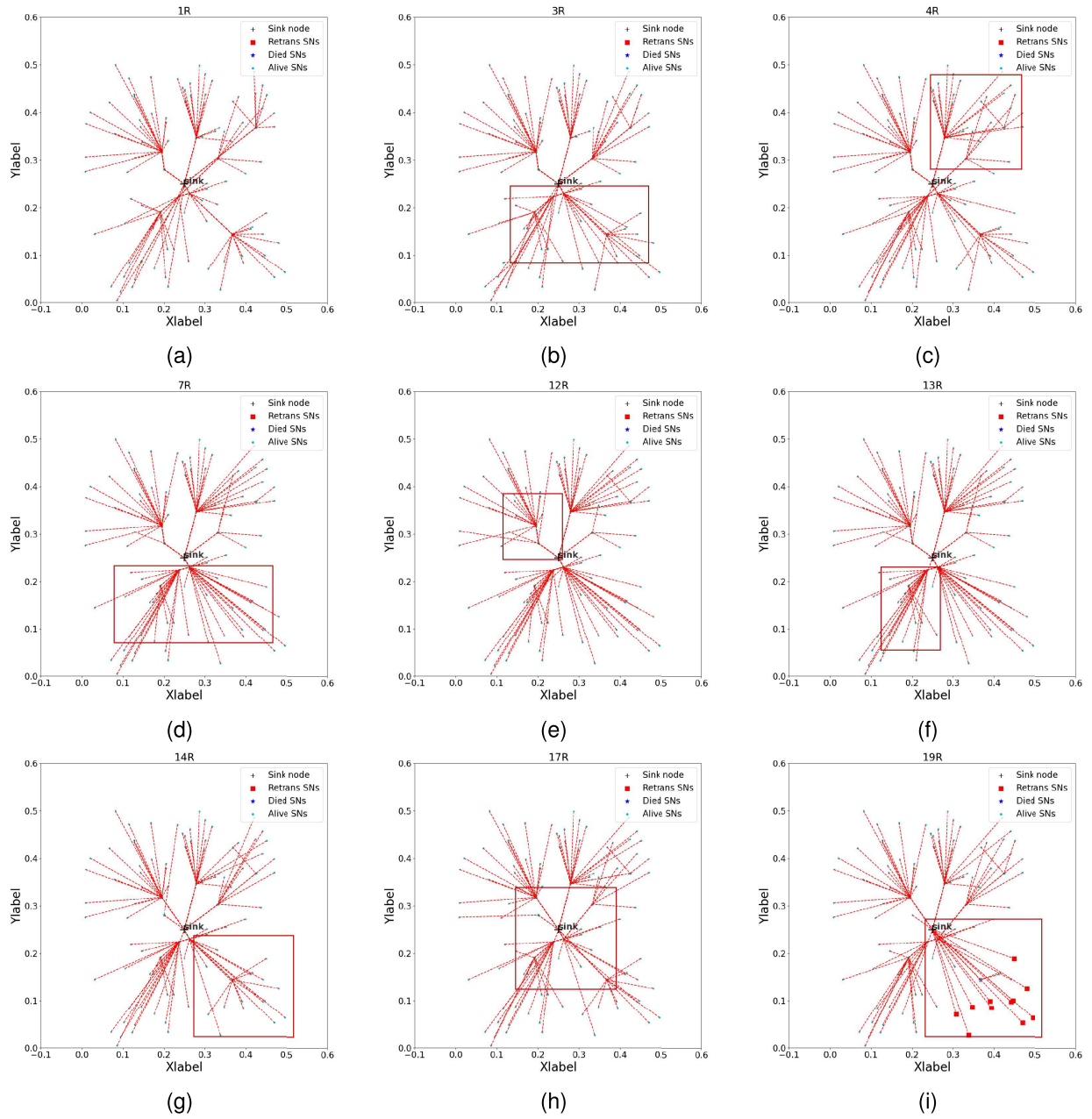


Fig. 8. MADII routing scheme when $\beta = 1\%$ (FND). (a) Round 1. (b) Round 3. (c) Round 4. (d) Round 7. (e) Round 12. (f) Round 13. (g) Round 14. (h) Round 17. (i) Round 19.

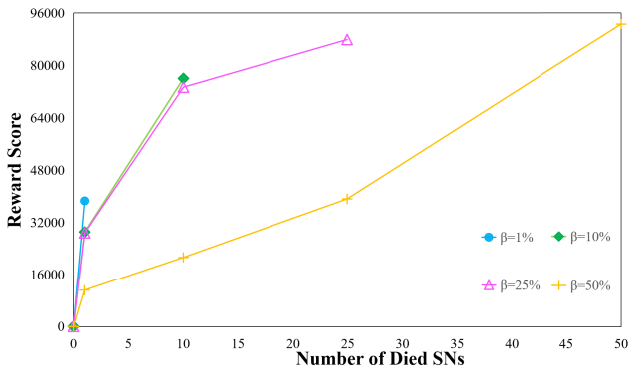


Fig. 9. RS with different network survival criteria.

fewer death SNs can learn more efficient routing scheme and get higher RSs when the same number of SNs die,

while the network survival criteria focusing on more death SNs need to balance the energy of the entire WSN nodes through the whole lifetime, and the efficient routing scheme cannot be carried out continuously similar to that of $\beta = 1\%$ (FND). This shows that different routing schemes should be adopted for the network survival criterion corresponding to specific goals.

V. CONCLUSION

In this article, we introduce an MADRL-based routing approach for WSN, named MADII, to optimize network performance. Our method employs an innovative transmission control mechanism STCIN3C that balances network load and designs a reward function considering the data transmission volume, data-receiving success rate, and the

relative energy consumption to promote the network efficiency. To address training convergence challenges and balance exploration–exploitation in MARL with large-scale action spaces, we propose an informer–IL hybrid policy that integrates IL with the informer’s ProbSparse attention mechanism. This approach dynamically adapts to WSN topology changes and mitigates the curse of dimensionality in MADRL. The extensive experimental results demonstrate that the proposed MADRL outperforms competing approaches in terms of EE and transmission efficiency for different network survival criteria. Future work will focus on integrating online routing techniques to address real-world deployment challenges, with emphasis on enhancing scalability and reliability under dynamic real-time environmental conditions.

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